The National Higher School of
Artificial Intelligence

ADVANCED DATABASES

Pr. Kamel BOUKHALFA

Slides From the Textbook :
Carlos Coronel and Steven Morris, Database Systems: Design, Implementation, and Management
Tenth Edition

Presentation

- Kamel Boukhalfa, Full Professor, National Higher School of Artificial Intelligence (2021)
- Head of the Research and Foresight Division at the National Sanitary Security Agency (since 2023)
- Full Professor, USTHB (2017)
- Phd From USTHB, Algiers, Algeria/ENSMA, Poitiers, France (2009)
- Teaching : Databases, Big Data, Datamining, C++ Programming, etc.
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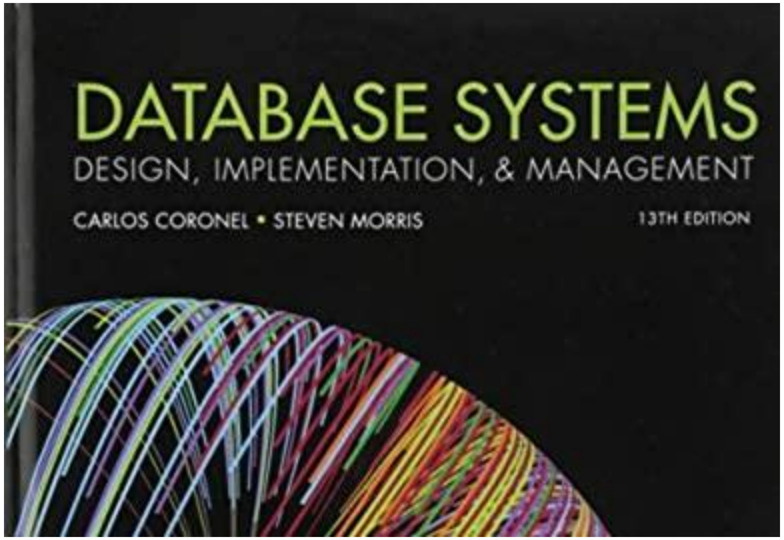
Information About the course

Course Description :

The objective of this course is to teach students advanced concepts of databases. Students will learn to optimise and tune databases before and after going into production. Students will learn how to design, manipulate and optimise distributed databases across a network. The course also focuses on specific kinds of databases such as data warehouses used for decision support as well as the new NoSQL databases used to overcome the scalability problems of relational databases. At the end of the course, the student will be able to easily administer and secure a database to protect it from the most dangerous threats.

Prerequisite : Databases, Operating Systems

Evaluation Method : Coursework (40%) + Final Exam (60%)



Course Content

- Database Performance Tuning and Query Optimization
- Distributed databases
- Business Intelligence and Data Warehouses
- NoSQL Databases
- Database Administration and Security

References

- Carlos Coronel and Steven Morris. Database Systems: Design, Implementation, & Management, 13th Edition, 2018.
- Raghu Ramakrishnan, Johannes Gehrke. Database Management Systems, McGraw-Hill Higher Education; 3rd edition. 2002
- Ramez Elmasri, Shamkant B. Navathe. Fundamentals of Database Systems, Pearson; 7th edition, 2015.

Evaluation

A) Course Test (30 mn) – 5%

B) Tutorial Test (60 mn) – 10%

C) LABS Test (60 mn) – 10%

D) Midterm Test (90 mn) - 15%

E) Final Exam (120 mn) : 60%

CHAPITRE 1

Database Performance Tuning and Query Optimization